



Internship Project Report: Fruit Quality AI

Project Title: Automated Fruit Freshness Detection using Deep Learning **Internship Domain:** Artificial Intelligence / Data Science **Project Status:** Completed & Verified

TEAM DETAILS:

- **Team ID:** LTVIP2026TMIDS91672
- **Team Leader:** Sankar Bendi
- **Team Members:**
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1. Ideation Phase

1.1 Problem Statement Manual inspection of fruit quality is a slow and labor-intensive process in the food industry. Traditional quality control relies on human sight, which is prone to fatigue and inconsistency. There is a growing need for automated systems in supermarkets and supply chains to accurately distinguish between fresh and rotten produce.

1.2 Proposed Solution Fruit Quality AI proposes an AI-powered classification system that leverages Transfer Learning to automatically classify fruit images into distinct categories such as:

- Fresh Apple / Rotten Apple
- Fresh Banana / Rotten Banana
- Fresh Orange / Rotten Orange The system aims to improve quality control efficiency, accuracy, and scalability.

1.3 Objectives

- Develop an automated fruit quality classification model.
- Utilize Transfer Learning (VGG16) for high-performance feature extraction.
- Achieve high classification accuracy (Targeting 97.4%).
- Build a responsive web interface for real-time inspections.

2. Requirement Analysis

2.1 Functional Requirements The system should:

- Accept fruit image uploads via a web interface.
- Preprocess images for model compatibility.
- Classify images into 6 distinct freshness categories.
- Display the final prediction result and uploaded image.

2.2 Non-Functional Requirements

- High prediction accuracy (97.4%).
- Production-ready user interface.
- Low latency for real-time uploads.
- Robust backend handling.

2.3 Hardware Requirements

- **Processor:** Intel i5 or higher.
- **RAM:** 8GB (minimum recommended).
- **Storage:** 5GB free space.

2.4 Software Requirements

- Python.
- TensorFlow / Keras.
- Flask (WSGI Application).
- VS Code / Jupyter Notebook.

3. Project Design Phase

3.1 System Architecture The Fruit Quality AI system consists of:

1. Dataset Module (13,599 images).
2. Preprocessing Module.
3. VGG16 Transfer Learning Model.
4. Flask Backend Engine.
5. HTML/CSS User Interface.

3.2 Data Flow

1. Image Input (Web Upload).
2. Image Preprocessing (Resizing/Normalization).
3. Feature Extraction (VGG16).
4. Classification Layer.
5. Prediction Output.

4. Project Planning Phase

4.1 Development Timeline | Phase | Description | | :--- | :--- | | **Ideation** | Problem identification & Domain research | | **Data Collection** | Gathering 10,901 training & 2,698 test images | | **Model Development** | VGG16 Transfer learning setup | | **Training & Testing** | 10 Epochs training & 97.4% validation | | **Deployment** | Flask integration & Localhost hosting |

5. Project Development Phase

5.1 Dataset Description The project utilizes a massive dataset of 13,599 images categorized into 6 classes: Fresh Apple, Fresh Banana, Fresh Orange, Rotten Apple, Rotten Banana, and Rotten Orange.

5.2 Data Preprocessing

- Image Resizing.
- Normalization.
- Data Augmentation.
- Train/Test Split.

5.3 Model Training

- **Approach:** Transfer Learning with VGG16.
- **Optimizer:** Adam.
- **Loss Function:** Categorical Crossentropy.
- **Accuracy:** 97.4%.

6. Project Documentation

6.1 Methodology Fruit Quality AI follows a Transfer Learning approach where a pre-trained VGG16 model is adapted for fruit classification.

1. Load VGG16 Pre-trained Weights.
2. Freeze base layers.
3. Add custom classification layers.
4. Train on fruit dataset.
5. Evaluate performance.

6.2 Challenges & Solutions

- **Challenge:** Dataset imbalance and model overfitting.
- **Solution:** Data augmentation and dropout regularization.

7. Project Demonstration

7.1 System Functionality The system features a Landing Page (index.html) and an Inspection Page (inner.html) where users can upload fruit images and receive an instant "Fresh" or "Rotten" status.

7.2 Results The trained model demonstrates a 97.4% validation accuracy with reliable predictions and a professional, responsive user interface.

Conclusion The project successfully demonstrates that Transfer Learning with VGG16 is highly effective for fruit quality detection. The final application is accurate, robust, and ready for deployment in real-world quality control environments